

# Find a Problem

## TODAY YOU'LL MAKE

**A list of problems worth fixing.**

### GET READY

- ☐ This book
- ☐ Pencil
- ☐ Two people to ask

### STEPS

- 1** Walk into every room in the house with this book.
- 2** Write down **10 things** that are annoying, slow, or easy to forget.
- 3** Ask two people what annoys them. Add theirs to the list.
- 4** Cross out any you can't build with what's on the shelf.
- 5** Circle your favourite **three**.

### IF IT BREAKS

**Can't think of anything?** Watch someone do a boring job. Wait for the sigh.

**Ideas too big?** Shrink them. Not a robot — a light that says the dog was fed.

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**YOUR CHALLENGE**

Pick the one that helps someone else, not you. Those are the best inventions.

**YOUR 10 PROBLEMS****I DID IT!**

Date: \_\_\_\_\_

**ONE THING I LEARNED**

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# Sketch the Fix

## TODAY YOU'LL MAKE

**Three drawings, and one winner.**

### GET READY

- ☐ This book
- ☐ Pencil
- ☐ Someone to show

### STEPS

- 1 Draw the problem at the top of the page.
- 2 Draw **three different** machines that fix it.
- 3 Label the parts of each: sensor, light, buzzer, motor.
- 4 Show all three to someone. Ask which they'd use.
- 5 Circle the winner. List every part you need.

### IF IT BREAKS

**Only one idea?** Draw the lazy version, the expensive version, and the tiny version.

**Can't draw?** Boxes and arrows count. Nobody is marking the picture.

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**YOUR CHALLENGE**

Draw the silliest one too. Sometimes that's the one that works.

**YOUR THREE IDEAS****I DID IT!**

Date: \_\_\_\_\_

**ONE THING I LEARNED**

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# Build the Ugly Version

## TODAY YOU'LL MAKE

**A rough build that actually works.**

### GET READY

- ☐ Laptop
- ☐ micro:bit
- ☐ Your parts list
- ☐ Cardboard + tape
- ☐ Pencil

### STEPS

- 1 Lay every part from your list out on the table.
- 2 Wire the sensor first. Get its number onto the screen.
- 3 Add the thing that happens — light, sound, or servo.
- 4 Hold it together with cardboard and tape. Don't make it pretty.
- 5 Make it work **once**, start to finish.

### IF IT BREAKS

**Everything broken at once?** Unplug all but the sensor. Add one part back at a time.

**Worked, now it doesn't?** A wire has moved. It is almost always a wire.

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**YOUR CHALLENGE**

Time yourself. Can you get from nothing to working in an hour? Ugly is allowed.

**DRAW HOW IT GOES TOGETHER**

**I DID IT!**

Date: \_\_\_\_\_

**ONE THING I LEARNED**

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# Test It on Someone

## TODAY YOU'LL MAKE

**One real fix, found by watching someone else.**

### GET READY

- ☐ Your invention
- ☐ Two testers
- ☐ Pencil

### STEPS

- 1 Hand it to someone. Say **nothing**.
- 2 Write down every place they get stuck.
- 3 Ask them: what did you think it would do?
- 4 Fix the **one** worst thing on your list.
- 5 Hand it to a second person. See if that thing is gone.

### IF IT BREAKS

**They didn't know what to do?** That is the result. Add a label, a light, or an instruction.

**It broke during the test?** Also a result. Tape the weak part and test it again.

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**YOUR CHALLENGE**

Test it on someone younger than you. They say what adults are too polite to say.

**WHERE THEY GOT STUCK****I DID IT!**

Date: \_\_\_\_\_

**ONE THING I LEARNED**

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# Invention Night

## TODAY YOU'LL MAKE

**Your invention, shown to a room full of people.**

### GET READY

- ☐ Your invention
- ☐ Spare batteries
- ☐ Card for a sign
- ☐ This book

### STEPS

- 1 Make a sign: the problem, your name, what it does.
- 2 Say your demo out loud **three times** on your own first.
- 3 Set it up on a table with spare batteries ready.
- 4 Invite everyone. Show it. Then let them try it.
- 5 Flip back through this book. All 25 weeks.

### IF IT BREAKS

**Won't work in front of people?** Fresh batteries, and a photo of it working as back-up.

**Nervous?** Say the problem first, then show it. Thirty seconds is plenty.

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**YOUR CHALLENGE**

Teach one person one thing you learned, well enough that they can do it without you.

**DESIGN YOUR SIGN****I DID IT!**

Date: \_\_\_\_\_

**ONE THING I LEARNED**

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